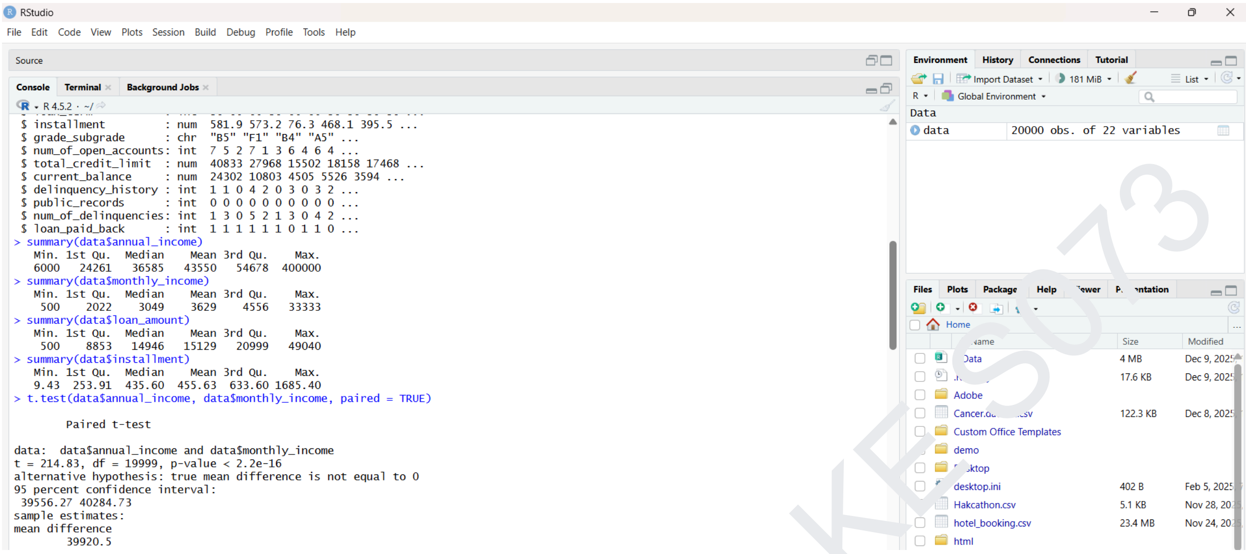
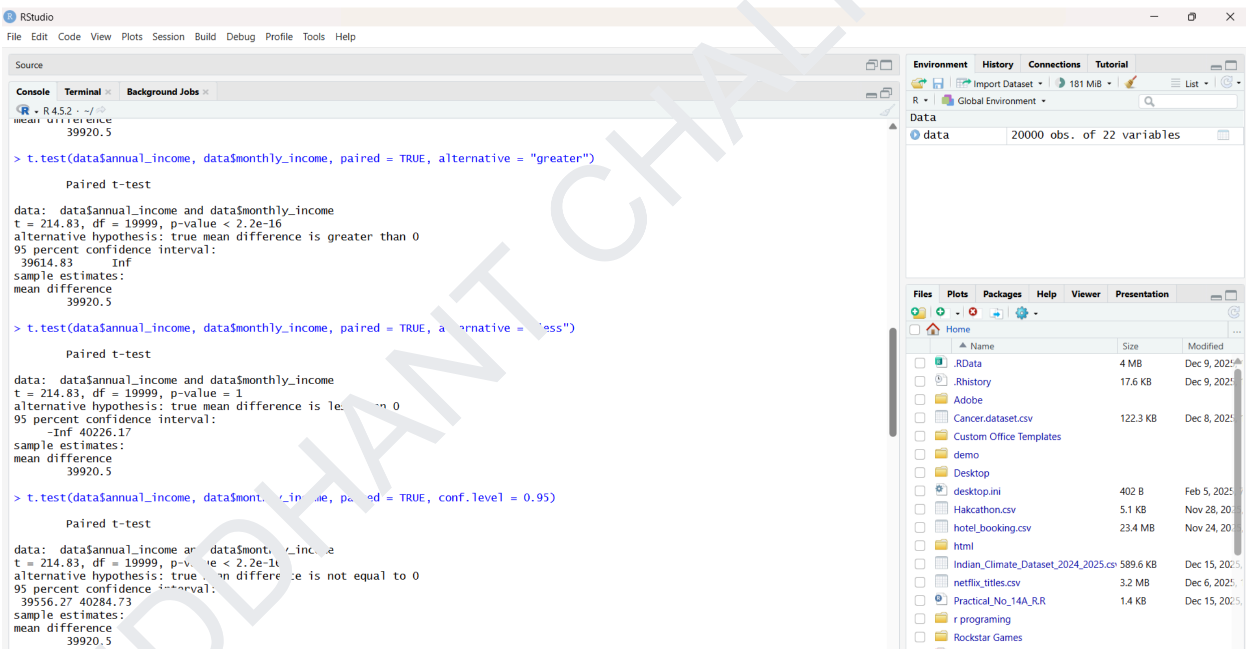
SHETH L.U.J. & SIR M.V. COLLEGE OF SCIENCE
SUBJECT - Data Analysis with R/SAS/SPSS

Aim - Performing paired t-tests using t.test(paired=TRUE) (R).

Output :





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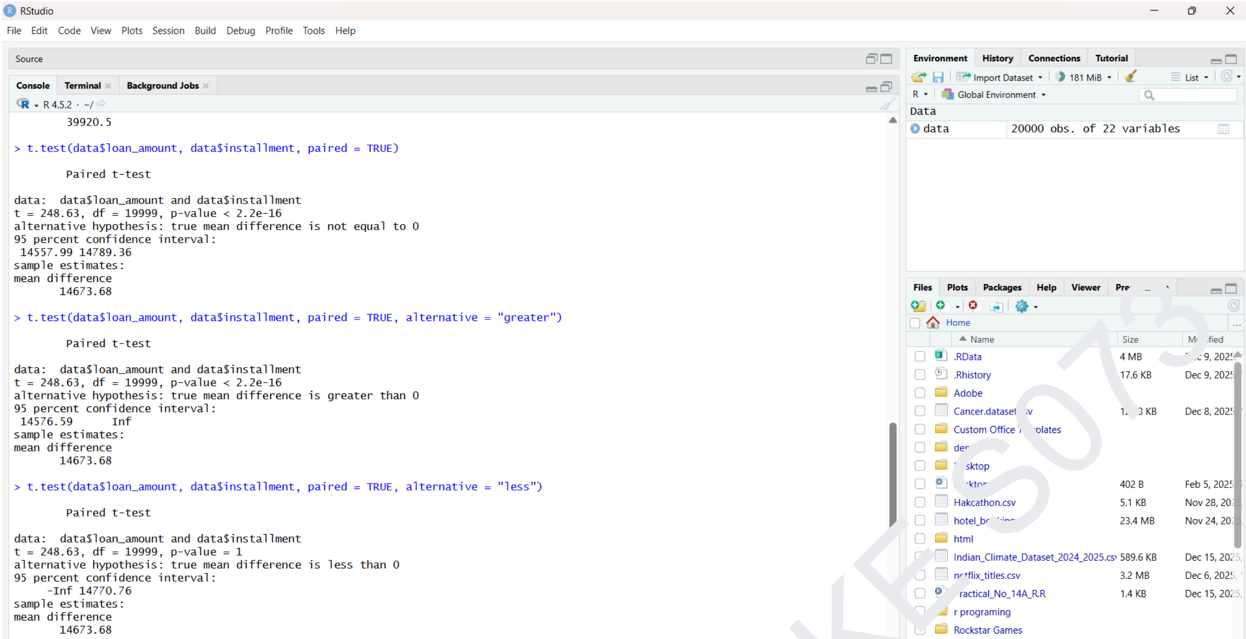
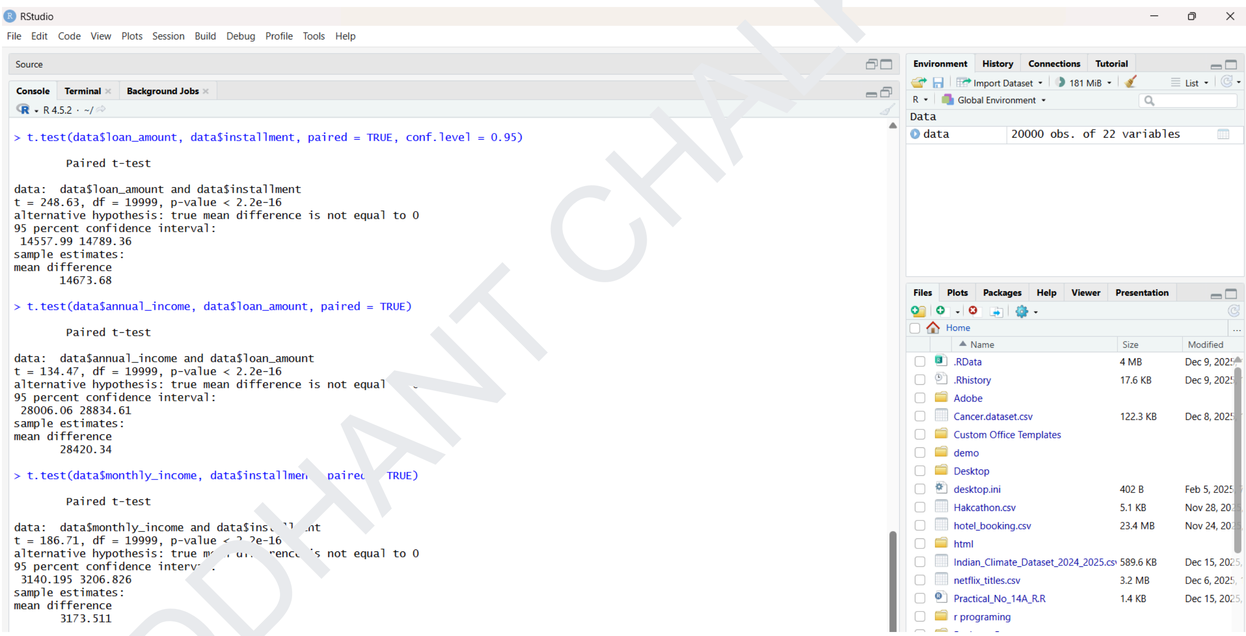


```
        39920.5

> t.test(data$loan_amount, data$installment, paired = TRUE)

	Paired t-test

data:  data$loan_amount and data$installment
t = 248.63, df = 19999, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 14557.99 14789.36
sample estimates:
mean difference 
       14673.68 

> t.test(data$loan_amount, data$installment, paired = TRUE, alternative = "greater")

	Paired t-test

data:  data$loan_amount and data$installment
t = 248.63, df = 19999, p-value < 2.2e-16
alternative hypothesis: true mean difference is greater than 0
95 percent confidence interval:
 14576.59      Inf
sample estimates:
mean difference 
       14673.68 

> t.test(data$loan_amount, data$installment, paired = TRUE, alternative = "less")

	Paired t-test

data:  data$loan_amount and data$installment
t = 248.63, df = 19999, p-value = 1
alternative hypothesis: true mean difference is less than 0
95 percent confidence interval:
     -Inf 14770.76
sample estimates:
mean difference 
       14673.68 
```



```
> t.test(data$loan_amount, data$installment, paired = TRUE, conf.level = 0.95)

	Paired t-test

data:  data$loan_amount and data$installment
t = 248.63, df = 19999, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 14557.99 14789.36
sample estimates:
mean difference 
       14673.68 

> t.test(data$annual_income, data$loan_amount, paired = TRUE)

	Paired t-test

data:  data$annual_income and data$loan_amount
t = 134.47, df = 19999, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 28006.06 28834.61
sample estimates:
mean difference 
       28420.34 

> t.test(data$monthly_income, data$installment, paired = TRUE)

	Paired t-test

data:  data$monthly_income and data$installment
t = 186.71, df = 19999, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 3140.195 3206.826
sample estimates:
mean difference 
       3173.511 
```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